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 calibration: {Gamma_1: 0.05, Gamma_2: 0.10, Gamma_3: 0.30, mod_1: 2.56, mod_2: 2.30, mod_3: 1.06}
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Abstract

We revise the F_U_Bi_i profile for TXS 0506+056 (the first identified neutrino blazar) using a 3-Gamma-point characterization: Gamma_1 = 0.05 THz (extreme flare, 2.56x), Gamma_2 = 0.10 THz (IceCube resonance, 2.30x), Gamma_3 = 0.30 THz (sustained emission, 1.06x). The monotonic decrease with increasing Gamma confirms thermal decoupling of the buoyancy-jet interaction at higher frequencies, with a gradient of 1.51x between extreme and sustained states.

1. Three-Point Characterization

Point	Gamma (THz)	Modulation	Physical State
1	0.05	2.56x	Extreme flare
2	0.10	2.30x	IceCube resonance
3	0.30	1.06x	Sustained emission

2. IceCube Resonance

At Gamma_2 = 0.10 THz, the modulation of 2.30x matches the target value of 2.3x (0.0% error), identifying this as the resonant frequency for neutrino-correlated buoyancy enhancement. This is where the IceCube-170922A neutrino event temporally coincided with the electromagnetic flare.

3. Monotonic Decrease

The gradient $\Delta_M / \Delta_{\text{Gamma}} = (2.56 - 1.06) / (0.30 - 0.05) = 6.0 / \text{THz}$ demonstrates that buoyancy modulation decays approximately linearly across the 3-point profile. The overall gradient ratio $2.56/1.06 = 1.51x$.

4. Implementation

File: fubi_i_txs0506_revised.py, classes TXS0506ExtremeFlareCalc, TXS0506IceCubeCalc, TXS0506SustainedEmissionCalc, TXS0506ThreeGammaProfileCalc. CP4 class #600. Tests: 8/8 pass.